LABORATORY PRACTICAL REPORT

of

Geographic Information System

(ICT. Ed. 473)

Submitted To

TRIBHUVAN UNIVERSITY

In Partial Fulfilment of the Requirements for the Course

BICTE – 7th Semester

Submitted By

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2082

CERTIFICATE

This is to certify that the Laboratory Practical Report of

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is a bona fide record of the experiments carried out by him under my guidance.

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Table of Contents

1. Explain Spatial Data and How to Think Spatially

Spatial data refers to any information that describes the position, shape, and location of objects on the Earth's surface. It helps identify where a feature exists and also provides additional details about it. Spatial data can appear in different forms such as points (representing locations like a tree, house, or electric pole), lines (showing features like roads, rivers, or boundaries), polygons (defining areas such as districts, lakes, or land parcels), and raster data (such as satellite images, elevation models, or temperature grids).

1.1 Types of Spatial Data

i. Vector Data

Vector data represents geographic features using precise geometric shapes such as points, lines, and polygons.

a) **Points** show the exact location of objects that have no measurable area (e.g., schools, trees, bus stops).
b) **Lines** represent features that have length but very small width compared to their length (e.g., roads, rivers, pipelines).
c) **Polygons** are used to represent enclosed areas or regions (e.g., countries, land parcels, lakes).

Vector data is highly accurate because it stores exact coordinates. It is ideal for mapping boundaries, transportation networks, administrative areas, and any feature where shape and position must be clearly defined.

ii. Raster Data

Raster data represents geographic information as a grid of cells or pixels, where each cell contains a specific value. This value usually represents a continuous variable. Examples include:

a) **Satellite imagery** (each pixel stores color or reflectance information)
b) **Elevation models (DEM)** (each pixel stores height)
c) **Temperature or rainfall maps** (each cell stores a climatic value)

Raster data is best suited for representing **continuous surfaces**, environmental patterns, and large-area phenomena that cannot be captured accurately as points or lines. Since raster is pixel-based, it is easier to perform mathematical operations and spatial analysis like slope, hillshade, NDVI, and classification.

1.2 How to Think Spatially

Spatial thinking involves using geographic perspective to understand how places, people, and objects interact across space. It requires not only knowing the location of features but also being able to interpret how those locations influence events and decisions. When thinking spatially, we examine how distance affects accessibility, how the arrangement of features creates patterns, and how movement or flow occurs between places. It also includes the ability to understand scale how something that appears small at the local level might become significant when viewed from a regional or global perspective. Spatial thinking encourages us to ask questions such as Why is this feature located here?, How does its position affect other features around it?, and What changes might occur if its location or environment changes?

By applying spatial thinking, we can interpret complex issues such as urban growth, disaster risk, environmental change, transportation planning, and population distribution. Instead of looking at information as isolated data points, spatial thinking allows us to connect them through maps, spatial models, and geospatial analysis. This approach helps reveal patterns and relationships that are not visible in traditional numerical tables or text-based data, making it an essential skill in geography, GIS, urban planning, environmental studies, and many scientific fields.

2. Steps to Install and Configure QGIS

2.1 Step-by-Step Installation

i. **Download QGIS Installer**

Go to the official QGIS website and download the installer for your OS (Windows, macOS, or Linux).

ii. **Run Installer**

Launch the downloaded installer, follow prompts, and install to default locations unless you need customization.

iii. **Complete Installation**

Once installed, QGIS appears as an application in your menu/desktop.

2.2 First Time Configuration

iv. **Open QGIS**

Start QGIS from the desktop shortcut or application menu.

v. **Enable Toolbars** (optional but useful)

In the View → Toolbars menu, tick relevant ones (Navigation, Digitizing, Map Tools, etc.) for easier workflow.

vi. **Set Project CRS (Coordinate Reference System)**

a) Go to Project → Properties → CRS

b) Choose a suitable system (e.g., EPSG:4326 WGS84).

This ensures your spatial data aligns correctly.

3. Steps to Design Maps Using QGIS

3.1 Basic Map Design Workflow

i. **Start New Project**

Project → New to open a blank QGIS project.

ii. **Add Data Layers**

a) Layer → Add Layer → Add Vector/Raster Layer

b) Browse and load your spatial files (shapefiles, GeoTIFF, etc.).

iii. **Arrange Layers**

In the **Layers Panel**, drag layers to control draw order.

iv. **Symbolize Layers**

a) Right-click a layer → Properties

b) Go to Symbology and choose styles (colours, categories, gradients) to represent data visually.

v. **Add Labels**

a) In Layer Properties → Labels, enable and style labels (names, values).

vi. **Map Layout for Printing**

a) Project → New Print Layout

b) Add map, legend, north arrow, scale bar, title.

c) Export as PDF/JPG/PNG.

4. How to Analyze Different Geographic Patterns

Geographic patterns describe the arrangement, distribution, and organization of features across a geographic space. To analyze these patterns, we look at how objects are spaced whether they appear clustered, randomly scattered, or evenly distributed. Understanding these patterns helps us interpret why certain phenomena occur in specific areas. For example, population may cluster around water sources, industries may concentrate along highways, and vegetation might vary with changes in elevation. Analysis often involves studying spatial relationships such as distance, density, direction, and connectivity between features. Tools like buffers, heat maps, spatial joins, and overlay analysis in GIS make it possible to identify hidden trends and relationships. By examining geographic patterns, we can better understand natural processes, human activities, resource distribution, environmental risks, and regional differences. This kind of analysis supports informed decision-making in fields such as urban planning, disaster management, transportation, environmental studies, and public health.

4.1 Common Spatial Patterns

i. **Clustered:** Clustered patterns occur when geographic features are located near each other in specific areas rather than being spread out evenly. This type of distribution often indicates a relationship between the features or a common reason for their location. For example, businesses may cluster in a city center where customers and resources are concentrated. Similarly, houses may cluster around water sources or transportation hubs. Analyzing clustered patterns helps identify hotspots and areas of intense activity, which is useful for planning services, infrastructure, or resource allocation.

ii. **Uniform (Even):** In a uniform or even pattern, features are spaced at relatively regular intervals across an area. This pattern often arises when resources or services are deliberately distributed to avoid overlap and ensure equal access. For example, trees planted in an orchard or street lights placed at regular distances along a road exhibit an even pattern. Uniform spacing can also result from natural competition between organisms, such as plants that require a minimum distance from one another for survival. Recognizing uniform patterns helps in understanding planning strategies and natural spacing mechanisms.

iii. **Random:** Random spatial patterns occur when features are distributed without any predictable order or clear arrangement. In this pattern, the location of each feature does not appear to depend on the position of other features. For example, the distribution of certain animal sightings or fallen leaves on a forest floor may be random. Random patterns suggest that there are few or no strong underlying spatial processes influencing where features occur. Identifying randomness is important because it indicates that spatial relationships such as clustering or uniformity are not present, which affects how we model and interpret the data.

4.2 Analysis Methods

i. **Proximity and Density:** Proximity analysis measures how close features are to each other in space. It helps determine relationships such as the distance between schools and hospitals or settlements and rivers. Tools like *buffers* create zones around features (e.g., a 1 km buffer) to show influence areas, accessibility, and impact zones.

   Density analysis examines how many features occur within a specific area. It identifies concentrations such as population clusters, traffic hotspots, or service gaps. Heat maps are commonly used to visualize density, making it easier to locate areas with high or low activity.

ii. **Overlay Analysis:** Overlay analysis involves placing multiple spatial layers on top of each other to identify relationships and interactions between them. For example, combining a population layer with a road network layer can show how population density relates to transportation access. Overlay analysis is powerful because it can reveal areas where conditions overlap, such as regions with both high hazard risk and high population. This method is widely used in environmental assessment, land use planning, and decision support systems.

iii. **Spatial Join:** A spatial join merges attributes from one layer to another based on their spatial relationships rather than common fields. For instance, if you have a layer of districts and a layer of schools, a spatial join can attach information about the number of schools to each district polygon. Spatial joins help summarize or integrate data across layers to create richer information. This method is essential for combining location-based attributes and performing meaningful analysis such as calculating total population per region or identifying which roads fall within a protected area.

5. Steps Using QGIS to Analyze Raster Data and Combine Raster & Vector Data

5.1 Raster Data Analysis in QGIS

i. **Load Raster Layer**
   Layer → Add Raster Layer and browse to your raster (e.g., elevation or imagery).

ii. **Open Raster Calculator**
   Raster → Raster Calculator to perform pixel-wise calculations (e.g., NDVI).

iii. **Clip Raster**
   Raster → Extraction → Clip Raster by Mask Layer to focus on area of interest.

iv. **Visual Analysis**
   Adjust **Symbology** (single band pseudo color, hillshade) to interpret terrain or features.

5.2 Combine Raster and Vector Data

i. **Add Both Raster + Vector Layers**
   Load your raster (e.g., DEM) and vector (e.g., administrative boundary).

ii. **Clip Raster Using Vector Polygon**
   a) Raster → Extraction → Clip Raster by Mask Layer
   b) Choose vector polygon as mask.

iii. **Zonal Statistics**
   a) Raster → Zonal Statistics
   b) Calculates stats (mean, sum, etc.) of raster values within each vector polygon.

iv. **Geoprocessing with Vector Based on Raster Results**
   a) e.g., create vector contours from DEM using Raster → Extraction → Contour.
   b) Use overlays or spatial joins to integrate raster-derived results with vector analysis.